I Earlier Fixed-Strength Calibration

Before the final calibration, KeyOpt fixed small strengths in advance and deployed the conditional residual only if an outer validation bound on support pairs passed. On ToolQA that bound removed the residual, so the configuration amounted to offsets plus event reading at a small fixed scale. This section collects every result of that configuration; none of them enters the main text, which uses the final calibration throughout. Table 25 gives its main table. On ToolQA the final calibration changes 132 GPT and 128 DeepSeek index choices; on AppWorld and ALFWorld (GPT-5.5) it changes none.

Held-out comparison of the two strength choices. On the 180 held-out tasks, the two configurations were compared directly (Table 26). With GPT-5.6-luna, fixed strengths complete 116 (index) and 120 (CompSkill) tasks against 119 and 124 for the final calibration; the index comparison favors the final calibration with both executors (6/3 and 2/0 rescues/harms) without reaching significance, as do CompSkill (5/1 and 0/0). Against the stronger baselines, the fixed-strength CompSkill exceeds SkillRet-8B (19/4, Holm $p = 0.010$) and BM25 (21/7, $p = 0.038$) but not Tool2Vec (8/13, $p = 0.77$) or RankGPT (6/4, $p = 0.77$). On the 332 evaluation tasks with GPT-5.6-luna it completes 251 against Tool2Vec's 237 (27/14, exact $p = 0.060$), versus 256 for the final calibration.

Rates, templates and cost. Table 32 adds failure reduction, tool calls and token cost for GPT-5.5 with the default CompSkill (264/332, 79.5%); without the reranker it completes 265/332. Of 30 AppWorld-C templates (all three tasks correct), the main-table calibrated index completes 25/19 with GPT/DeepSeek. The template counts refer to the same index-only configurations as Table 25B.

ToolQA components. The CompSkill row in Table 25A uses the default scorer $S = r_t(s) + q_t^\top \Delta k_s + b_s$ with the validation-selected residual; the component ablations in Table 28 share its index. The reranker-free variant uses $S = q_t^\top \Delta k_s + b_s$ with a bounded residual ($\|\Delta k_s\|_2 \le 0.6$ for GPT-5.5 and 0.3 for DeepSeek); see Table 41.

Configuration scope. On ToolQA and ALFWorld, the main-table CompSkill retains the frozen reranker, KeyOpt, hypergraph and review; on AppWorld, whose Challenge apps never occur in support, it reports the calibrated index (Appendix I.2). Separate AppWorld Full runs add a dense Top-1 fallback. In Table 37, the Claude column uses the group-evidence variant over GoSkills bundles without evidence recall or review, and the 128-success DeepSeek group-evidence variant uses an operation scaffold; DeepSeek GoSkills retains its 16,384-token reply contract. SRA-specific and SkillRet seed-17 ToolQA fits share Qwen-collected feedback across executors, one run each.

I.1 Development Exposure and Configuration Selection

ToolQA's 332 evaluation tasks, the 90 AppWorld Challenge tasks and the 134 ALFWorld `valid_unseen` games served as development pools; when this configuration was fixed, no separate held-out execution test existed (Appendix H.1 adds two for the final calibration). Table 29 lists every configuration decision, the alternatives that were evaluated and whether evaluation outcomes had been observed when it was fixed.

Two facts limit the resulting selection effect. First, compressor training and the SkillRet comparison never used execution outcomes. Second, the ToolQA conclusion holds for all three complete configurations, so it does not depend on which of them is reported (Table 30 lists all three).

I.2 Benchmark-Specific Configurations

Full denotes the complete pipeline: compact keys, KeyOpt, hypergraph recall and cited review. Table 25 reports Full on ToolQA and ALFWorld, retaining the reranker, KeyOpt, hypergraph and reviewer. For AppWorld's app-specific API combinations and disjoint support/test apps, we use the calibrated index without history-based recall or review. These settings were chosen on the public development tasks rather than a separate validation set. Baselines remain at their stated reference settings. The reranker-free ToolQA result (265 with GPT-5.5) remains an ablation outside the main table. Separate AppWorld Full runs use the frozen checkpoint together with a dense Top-1 fallback and a different candidate protocol.

Evidence scope. The ToolQA ablations, costs and paired tests in this appendix use the reference architecture of Table 25; AppWorld reports the index-only configuration (81/74), and other ALFWorld variants are listed in Table 37.

I.3 Main-Table Metrics with Fixed Strengths

Table 32 reports the original execution records: Vector Top-4 has 256 successes here, versus 258 in

(A) ToolQA ($n = 332$; 178 complex queries; 785 documents)

		DeepSeek					GPT-5.5				
Method	Docs	Succ. (%)	Cmp.	Worst	Hit@1	R/H	Succ. (%)	Cmp.	Worst	Hit@1	R/H
No Skills	0	200 (60.2)	37.1	41.4	—	72/13	215 (64.8)	41.6	41.4	—	58/9
Vector Top-4	4	240 (72.3)	59.0	51.4	63.0	33/14	258 (77.7)	64.6	67.1	63.0	17/11
Vector Top-1	1	236 (71.1)	57.3	55.7	63.0	27/4	248 (74.7)	60.7	61.4	63.0	21/5
GoSkills	2.57	218 (65.7)	43.8	53.3	44.9	57/16	241 (72.6)	53.4	65.2	44.9	38/15
SkillRouter	1	226 (68.1)	52.8	51.4	46.7	40/7	238 (71.7)	54.5	51.4	46.7	30/4
CompSkill	1	**259 (78.0)**	**66.3**	**64.3**	**78.3**	—	**264 (79.5)**	**68.0**	**68.6**	**79.8**	—
Δ over best baseline	—	**+19 (+5.7)**	**+7.3**	**+8.6**	**+15.4**	—	**+6 (+1.8)**	**+3.4**	**+1.4**	**+16.9**	—

	(B) AppWorld Challenge ($n = 90$; 469 APIs)				**(C) ALFWorld** ($n = 134$ `valid_unseen`; 37 skills)	
	DeepSeek		GPT-5.5		DeepSeek	GPT-5.5
Method	TGC	SGC	TGC	SGC	Succ. (%)	Succ. (%)
No Skills	68 (75.6)	14	78 (86.7)	22	112 (83.6)	125 (93.3)
Vector Top-4	70 (77.8)	14	78 (86.7)	23	125 (93.3)	130 (97.0)
Vector Top-1	69 (76.7)	18	**81 (90.0)**	**25**	126 (94.0)	128 (95.5)
GoSkills	**74 (82.2)**	**20**	**81 (90.0)**	**25**	126 (94.0)	130 (97.0)
SkillRouter	65 (72.2)	16	80 (88.9)	23	126 (94.0)	130 (97.0)
CompSkill	**74 (82.2)**	19	**81 (90.0)**	**25**	**127 (94.8)**	**131 (97.8)**

DeepSeek output-limit cases use a 65,536-token reply budget (Appendix I.5); unknowns stay in n. Δ: gain over the best baseline in each column. R/H: known-pair rescues/harms against the row. AppWorld uses the calibrated index, without hypergraph recall or review. Configurations: Appendix I.2. GoSkills uses full documents (Appendix I.5). —: not applicable or not reported.

Table 25: Main table under the earlier fixed-strength calibration, with GPT-5.5 and the September DeepSeek checkpoint. AppWorld and the GPT-5.5 ALFWorld column coincide with Table 3, because the final calibration changes none of their selections. (A) ToolQA success, complex queries (Cmp.), worst domain (Worst) and first-choice Hit@1. (B) AppWorld task/template completion. (C) ALFWorld success.

Method	Docs	GPT-5.6-luna	DeepSeek	
		n=180	n=100	n=180, all
Vector Top-1	1	106	64	115
SkillRouter	1	106	61	110
Vector Top-4	4	126	70	125
Fixed: index	1	116	66	124
Final: index	1	119	68	128
Fixed: CompSkill	1	**120**	**70**	**130**
Final: CompSkill	1	**124**	**70**	**130**

Table 26: Fixed versus final strengths on the 180 held-out ToolQA tasks with both executors (September DeepSeek checkpoint). GPT-5.6-luna executes with the unchanged GPT-5.5 fits. The DeepSeek column marked all includes 80 tasks seen by an earlier DeepSeek study; the main DeepSeek cohort (n=100) excludes them. Unknown outcomes (at most three) count as failures.

Comparison	Cohort	R/H	p
Fixed vs. Vector Top-1	GPT, 180	19/5	0.007
Fixed vs. SkillRouter	GPT, 180	19/5	0.007
Fixed vs. Vector Top-1	DS, 100	8/2	0.109
Fixed vs. SkillRouter	DS, 100	10/1	0.012
Fixed vs. Vector Top-1	DS, 180	18/3	0.002
Fixed vs. SkillRouter	DS, 180	22/2	<0.001
Final Fixed vs. Vector Top-1	GPT, 180	22/4	<0.001
Final Fixed vs. SkillRouter	GPT, 180	23/5	<0.001
Fixed vs. Vector Top-4	GPT, 180	8/14	0.286
Fixed vs. Vector Top-4	DS, 100	3/2	1.000

Table 27: Held-out paired comparisons of fixed-strength and final CompSkill (rescues/harms, exact two-sided McNemar on known pairs). GPT: GPT-5.6-luna; DS: September DeepSeek checkpoint.

Table 25. Tool-call and token figures retain their original scope. CompSkill uses 2.88 tool calls per episode versus 3.39 for GoSkills in these records.

I.4 Results by Domain and Query Type

Table 33 and Figure 8 break ToolQA success down by data domain and query type. CompSkill has the highest complex-query success for both executors and the highest Flights success, the domain with the most confusable database guides. Coffee is near ceiling for every selector that supplies the Coffee guide, and Yelp separates selectors less than Flights and Airbnb. CompSkill's gains therefore come from the domains where the right procedure is hardest to recognize from its description.

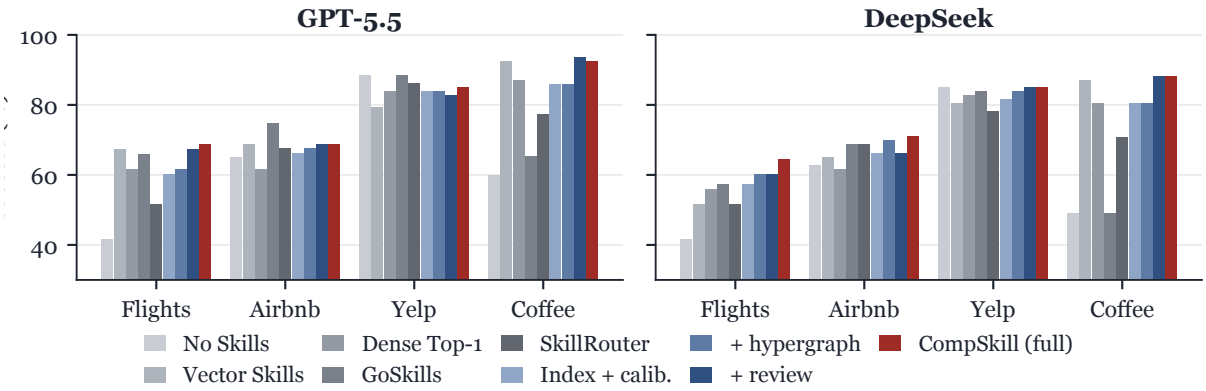


Figure 8: ToolQA success by data domain for every selector and CompSkill variant.

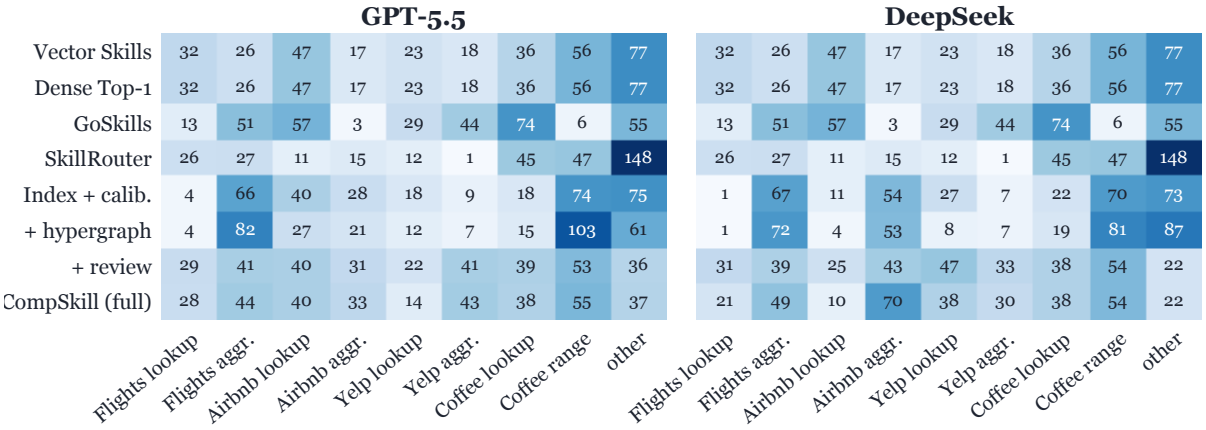


Figure 9: Which document each selector supplies first on 332 ToolQA tasks: the eight database guides (lookup, aggregation or range variants per domain) and all other skills, pooled in one column.

Component	Without	With
Compact keys (SkillRet NDCG@10)	62.2	**70.9**
KeyOpt offsets (GPT / DS)	241 / 232	**249 / 240**
+ residual, own feedback (GPT)	249	**257**
+ residual, Qwen (GPT / DS)	246 / 239	**255 / 247**
Cited review, no events (GPT / DS)	249 / 240	**262 / 252**
+ events (GPT / DS)	262 / 252	**264 / 259**

Table 28: Each component versus its matched reference (keys: truncated 512-d; others: ToolQA successes/332). All residual rows are significant ($p < 0.05$; Section 4.6).

Table 34 and Figure 9 show what reaches the executor. Most of the 785 documents are web skills that describe the same services (airline sites, booking platforms, coffee retailers), and SkillRouter supplies one of them in 44.6% of tasks. CompSkill supplies a guide of the task's own data domain in 88.0% of tasks with GPT-5.5 (292/332) and 93.4% with DeepSeek (310/332), more than any baseline listed here. Within a domain, different procedural guides can still produce different outcomes. Appendix I.11 separates these switches from changes that introduce a missing domain guide.

I.5 Completed Full-Document Comparisons

Full-document GoSkills. The earlier ToolQA GoSkills renderer capped each document at 1,800 characters and total context at 9,000. The completed adaptation restores full documents while retaining group members, their order, roles and execution contract. Table 25 reports 241/91/0 and 218/114/0 success/failure/unknown counts (GPT/DeepSeek), replacing the capped 245/215 successes. Selected DeepSeek cases use a 65,536-token reply limit and a 192,000-token cumulative pre-request output gate; the default is 16,384/24,000, with 20 steps and 900 seconds in both profiles. Against the unchanged main-table Full, 264/259, rescues/harms are 38/15 and 57/16. Complex-query rates are 53.4/43.8%, worst-domain rates 65.2/53.3%, and mean document

Decision	Alternatives evaluated (where reported)	Outcomes seen first?
Compressor architecture, epochs, seed	SkillRet TRAIN internal split only (Appendix F.3)	No
SkillRet methods and seeds	fixed before first TEST access; later controls marked †	No (controls: yes)
Splits, executors, action/time budgets	fixed protocol; one DeepSeek output-limit re-execution (Appendix I.5)	No
Baseline settings	released settings; GoSkills full-document rendering (Appendix I.5)	Yes (GoSkills)
Reading, recall and review hyperparameters	set during development, then fixed across benchmarks and executors	Yes
Reranker in the default scorer	with and without reranker (Table 41)	Yes
Residual deployment	support-validated gate (default); fixed scales $\times 1$, $\times 2$ (Table 41)	Gate: no; scales: yes
Review citation interface	exact quote (default); event-row IDs (Table 35)	Yes
AppWorld configuration	calibrated index (main); Full runs (Appendix F.2)	Yes
ALFWorld configuration	tie order and Full variants (Table 37)	Yes

Table 29: Development exposure of the earlier configuration: each configuration decision, the alternatives evaluated, and whether evaluation outcomes had been observed before it was fixed.

Complete ToolQA configuration	GPT	DS
Default: reranker, validated KeyOpt, exact quote	264	259
Without reranker, residual $\times 2$	265	256
With reranker, event-row citation IDs	265	261 (1)
Strongest baseline (Vector Top-4)	258	240

Table 30: All three complete configurations exceed the strongest ToolQA baseline, Vector Top-4, for both executors (successes of 332, unknowns in parentheses).

count 2.57. Earlier domain plots, cost summaries and bootstrap tables retain the capped reference.

Citation-interface Full. Table 35 retains the original reranker, KeyOpt and event history, changing the review citation interface to explicit event-row IDs. Its six baseline contrasts form one Holm family. Against full-document GoSkills, rescues/harms are 38/14 and 50/7 in the completed table, net gains of 24 and 43–44 tasks. For the original execution records only, the saved 95% bootstrap envelopes are $[2.71, 13.55]$ and $[8.13, 18.98]$ percentage points, and Holm-adjusted known-pair p values are 0.0028 and 1.09×10^{-8}. These statistics apply to the original rows only. All 332 source groups are singletons; intervals are task-level. Component ablations retain their original records.

I.6 Paired Tests and Intervals

Table 36 gives task-level bootstrap intervals for CompSkill against each reference baseline (GoSkills with capped rendering; the full-document GoSkills comparison is in Appendix I.5), and Figure 10 shows the full matrix of paired net gains. With DeepSeek the interval excludes zero against every baseline; with GPT-5.5 it excludes zero against every single-document baseline. Every selector loses net tasks to CompSkill.

I.7 AppWorld and ALFWorld

On AppWorld Challenge, the calibrated index in Table 25B completes 81/90 (GPT-5.5) and 74/90 (DeepSeek). Figure 11 resolves its task and template completion. Figure 11 resolves the AppWorld Challenge results by template. Challenge templates use apps absent from support, so no same-app events are available. The figure retains the original GPT No Skills record: 77 successful tasks and 21 complete templates. The main table reports 78 tasks and 22 templates under completed outcome accounting. The calibrated index completes 81 tasks and 25 templates, versus 22 for No Skills; DeepSeek completes 19 templates.

In this earlier configuration, CompSkill completes 131/134 ALFWorld games with GPT-5.5 and 127/134 with DeepSeek (September checkpoint, original tie order). Table 37 lists every evaluated configuration with its executor-specific settings.

I.8 Calibration Analyses with Fixed Strengths

Table 38 details the evolution study of Figure 12b. Each stage refits on one, two or three thirds of the same 318-task support set while keeping the evaluation tasks and residual scale fixed. The table reports every stage for both executors with paired rescues/harms and unknown counts.

Table 39A crosses indexing and calibration choices under transferred Qwen3-8B feedback; panel B gives the GPT-5.5 component path with the frozen reranker. Figure 13 shows the paired rescues and harms of every nonzero KeyOpt fit.

Benchmark	Executor	Success	Configuration reported in Table 25
ToolQA	GPT-5.5	264/332	Reference reranker, validation-selected KeyOpt (offset level), event recall and document/event review.
ToolQA	DeepSeek	259/332	Reference nine-arm Full with its own support feedback.
AppWorld	GPT-5.5	81/90	Compact index, frozen reranker and KeyOpt; no hypergraph recall or review; 25/30 complete templates.
AppWorld	DeepSeek	74/90	Same index-only configuration with executor-specific calibration; 19/30 complete templates.
ALFWorld	GPT-5.5	131/134	Semantic tie ordering and original Full review.
ALFWorld	DeepSeek	127/134	Original Full with the legacy tie order.

Table 31: Main-table configurations of the earlier fixed-strength calibration. AppWorld uses the calibrated index; ToolQA and ALFWorld use Full. Each result keeps its benchmark-specific adaptations.

Method	Succ.	Err.↓	Complex	Worst	Hit@1	Docs	Tools	Actor out	In
No Skills	215	41.9	41.6	41.4	—	0	3.83	350	28.8
Vector Top-4	256	10.5	64.0	67.1	63.0	4	2.89	245	43.2
GoSkills	245	21.8	55.6	65.2	44.9	2.55	3.39	323	39.8
SkillRouter	238	27.7	54.5	51.4	46.7	1	3.19	279	29.3
CompSkill	**264**	—	**68.0**	**68.6**	**79.8**	1	**2.88**	242	40.1

Table 32: Original ToolQA GPT-5.5 execution records ($n = 332$). Err.↓: reduction in non-successes; Complex/Worst: success rates; Tools/Actor out: per execution; In: actor plus reviewer input tokens per task (K). Reviewer output is reported separately in Table 43. Costs cover these records; support collection and fitting are one-off costs.

Method	Flights (70)	Airbnb (83)	Yelp (87)	Coffee (92)	Lookup (154)	Complex (178)
GPT-5.5						
No Skills	41.4	65.1	**88.5**	59.8	91.6	41.6
Vector Top-4	67.1	68.7	79.3	92.4	92.9	64.6
Vector Top-1	61.4	61.4	83.9	87.0	90.9	60.1
GoSkills	65.7	**74.7**	**88.5**	65.2	**94.8**	55.6
SkillRouter	51.4	67.5	86.2	77.2	91.6	54.5
Index + calib.	60.0	66.3	83.9	85.9	88.3	63.5
+ hypergraph	61.4	67.5	83.9	85.9	88.3	64.6
+ review	67.1	68.7	82.8	**93.5**	93.5	66.3
CompSkill (full)	**68.6**	68.7	85.1	92.4	92.9	**68.0**
DeepSeek						
No Skills	41.4	62.7	**85.1**	48.9	87.0	37.1
Vector Top-4	51.4	65.1	80.5	87.0	87.7	59.0
Vector Top-1	55.7	61.4	82.8	80.4	87.0	57.3
GoSkills	57.1	68.7	83.9	48.9	90.3	42.7
SkillRouter	51.4	68.7	78.2	70.7	85.7	52.8
Index + calib.	57.1	66.3	81.6	80.4	85.1	61.2
+ hypergraph	60.0	69.9	83.9	80.4	86.4	64.0
+ review	60.0	66.3	**85.1**	**88.0**	89.0	64.6
CompSkill (full)	**64.3**	**71.1**	**85.1**	**88.0**	**91.6**	**66.3**

Table 33: ToolQA success (%) under the earlier fixed-strength calibration, by data domain and query type (number of tasks in parentheses), with each executor's own feedback. Bold: best per column.

Frozen calibration transfer. Calibration fitted on one executor transfers to another without refitting: DeepSeek's calibration raises GPT-5.5 from 241 to 252/332 successes, and GPT's raises DeepSeek from 232 to 238 in the completed accounting (the original records give 230 to 236); calibration fitted on Qwen3-8B feedback adds up to 9 further successes ($p < 0.05$). In execution, compact keys alone match dense retrieval (232 versus 234 for DeepSeek, 244 versus 245 for GPT-5.5) and exceed it by 29 tasks for Qwen3-8B; with Qwen3-8B on ALFWorld, CompSkill beats No Skills, SkillRouter and Vector Top-4 (50 versus 41, 40, 37). We transfer the already fitted GPT-5.5 and DeepSeek checkpoints between executors, holding the compact Top-10, frozen reranker, original documents and within-executor budgets fixed. The no-calibration row uses compact retrieval plus reranking, not No Skills or full-library SkillRouter. Neither source is refitted on target feedback. The

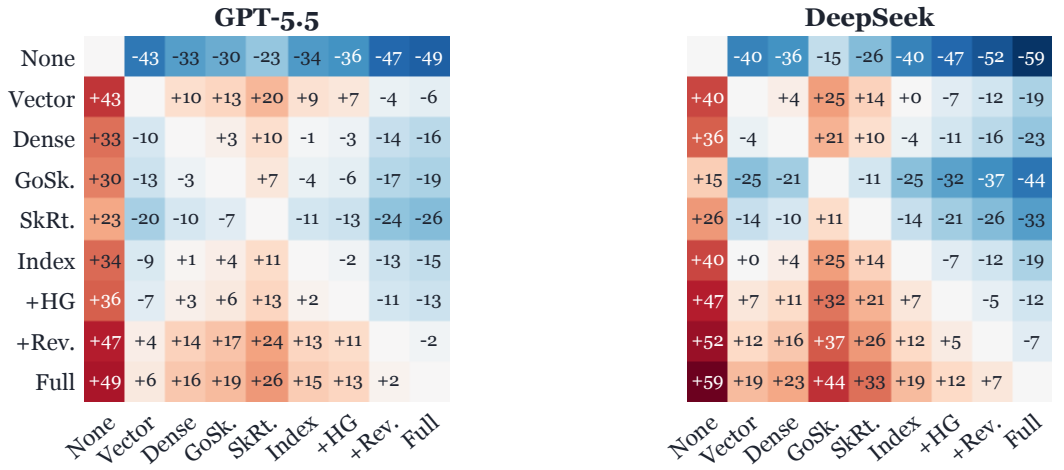


Figure 10: Net paired gain (rescues minus harms) of the row selector over the column selector on 332 ToolQA tasks.

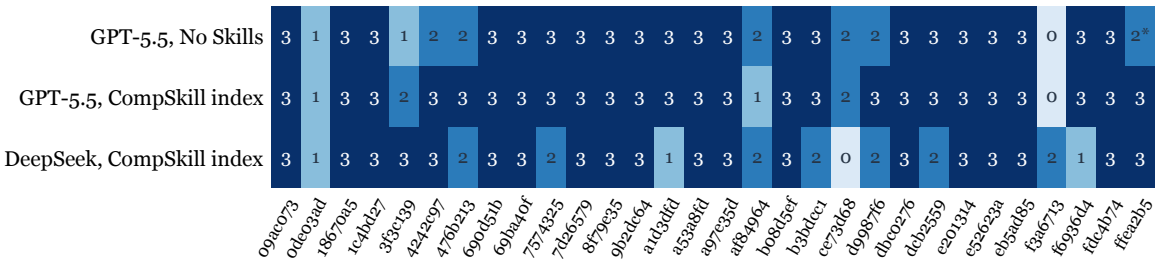


Figure 11: AppWorld index-only comparison by Challenge template (successes out of the three tasks per template); * marks an original unknown outcome. Table 25B uses completed outcome accounting.

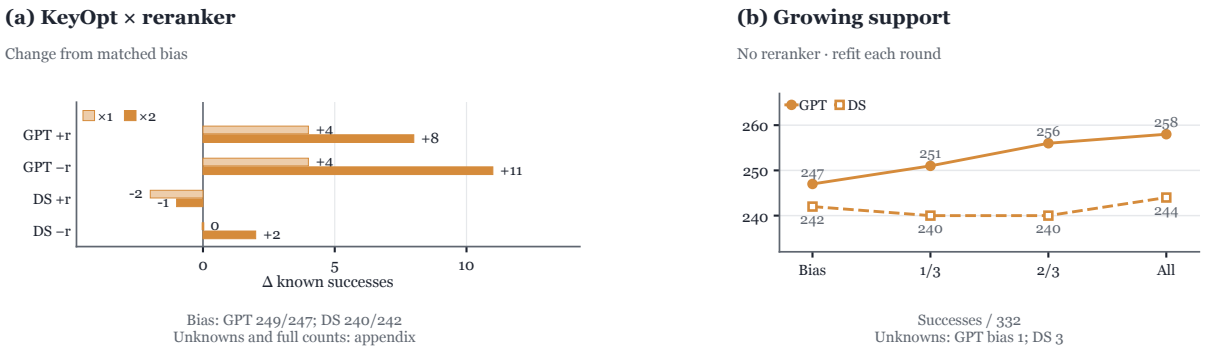


Figure 12: KeyOpt detail, original executions: (a) residual scales 1 and 2 crossed with reranking; (b) refitting on one, two or three thirds of the 318 support tasks, evaluated on all 332 ToolQA tasks.

six rows comprise 1,992 outcomes from 990 executions, of which 754 are shared with earlier runs.

Source evidence and parameters. Both sources planned the same 318 support tasks with two matched documents per task. GPT has 633 known and three unknown arms; DeepSeek has 626 known and ten unknown, with 623 arms known to both. Both validation fits select offset-level calibration, so the transferred state consists of reliability offsets (12 GPT keys and 16 DeepSeek keys). This study therefore evaluates frozen offset calibration; Appendix C covers residual fits.

Paired comparisons, original executions. The four paired tests compare own-source calibration against no calibration and against the other source. For GPT, own versus none rescues 19 tasks and harms 11 (net $+8$); own versus DeepSeek rescues zero and harms three (net -3). For DeepSeek, own versus none gives 17/9 rescues/harms on 329 known pairs, with full-denominator net bounds

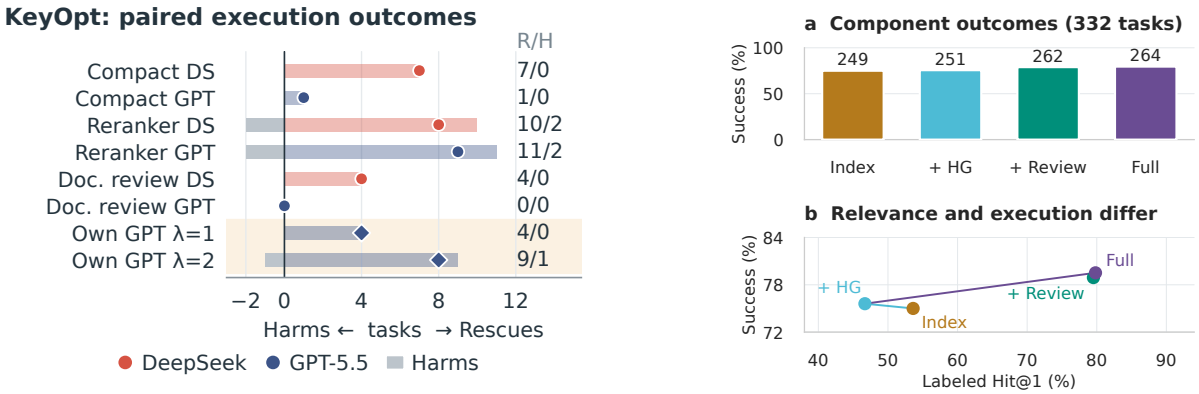


Figure 13: Left: KeyOpt rescues/harms against bias-only calibration (shaded: GPT-5.5 own feedback). Right: GPT-5.5 component outcomes and relevance versus success on all 332 ToolQA tasks.

First document	Guide	Domain	Label	Other
GPT-5.5				
Vector Top-4	76.8	76.8	63.0	23.2
Vector Top-1	76.8	76.8	63.0	23.2
GoSkills	83.4	82.2	44.9	16.6
SkillRouter	55.4	55.4	46.7	44.6
Index + calib.	77.4	77.4	53.6	22.6
+ hypergraph	81.6	67.8	46.7	18.4
+ review	89.2	89.2	79.5	10.8
CompSkill (full)	88.9	88.0	79.8	11.1
DeepSeek				
Vector Top-4	76.8	76.8	63.0	23.2
Vector Top-1	76.8	76.8	63.0	23.2
GoSkills	83.4	82.2	44.9	16.6
SkillRouter	55.4	55.4	46.7	44.6
Index + calib.	78.0	78.0	53.6	22.0
+ hypergraph	73.8	70.2	45.8	26.2
+ review	93.4	93.4	83.7	6.6
CompSkill (full)	93.4	93.4	78.3	6.6

Table 34: What each selector supplies first on ToolQA (% of 332 tasks): any of the eight database guides (Guide), a guide of the task's data domain (Domain), the labeled guide (Label), or a public skill (Other).

Completed comparison	GPT-5.5	DeepSeek
Dense Top-1	249 (0)	235 (4)
SkillRouter	238 (0)	225 (2)
GoSkills, full documents	241 (0)	218 (0)
Full, citation-ID interface	265 (0)	261 (1)

Table 35: Full-document comparison: successes/332 (unknowns). Budgets are specified above; Citation-ID Full uses a review profile of its own.

$[7, 8]$; own versus GPT gives 6/4, with bounds $[1, 2]$. Holm-corrected exact McNemar p-values are 0.675, 0.675, 0.675 and 0.754, respectively.

Interaction and scope, original executions. The difference between the GPT- versus DeepSeek-source effect under the two executors is -0.30

Against	Succ.	Δ	95% CI	R/H	p
GPT-5.5					
No Skills	215	+49	[+34, +64]	58/9	$< 10^{-4}$
Vector Top-4	258	+6	[-4, +17]	17/11	0.34
Vector Top-1	247	+17	[+7, +27]	21/5	0.002
GoSkills	245	+19	[+6, +33]	34/15	0.009
SkillRouter	238	+26	[+15, +37]	30/4	$< 10^{-4}$
Index + calib.	249	+15	[+7, +24]	17/2	0.001
+ hypergraph	251	+13	[+5, +21]	15/2	0.002
+ review	262	+2	[-3, +8]	5/3	0.73
DeepSeek					
No Skills	200	+59	[+42, +76]	72/13	$< 10^{-4}$
Vector Top-4	240	+19	[+6, +32]	33/14	0.008
Vector Top-1	236	+23	[+13, +34]	27/4	$< 10^{-4}$
GoSkills	215	+44	[+28, +60]	58/14	$< 10^{-4}$
SkillRouter	226	+33	[+20, +46]	40/7	$< 10^{-4}$
Index + calib.	240	+19	[+9, +30]	24/5	0.001
+ hypergraph	247	+12	[+2, +22]	19/7	0.029
+ review	252	+7	[+1, +14]	9/2	0.065

Table 36: CompSkill against every other row on ToolQA: the row's successes, CompSkill's difference, task-level paired bootstrap interval (10,000 draws, seed 17), rescues/harms and exact McNemar p.

ALFWorld	GPT	DS	Qwen	Claude
No Skills	125	112	41	118
Vector Top-4	130	125	37	—
Vector Top-1	128	126	—	—
GoSkills	130	126	56	124[1]
SkillRouter	130	126	40	120[2]
SkillNet-Fabric	129	127	63	124
Group evidence	130	128	—	126
+ hypergraph, no review	130	124	50	—
CompSkill, legacy tie order	129	127	50	—
CompSkill, main table	**131**	127	—	—

Table 37: ALFWorld successes (of 134) by CompSkill configuration and executor. The DeepSeek group-evidence result (128) uses an operation scaffold.

percentage points on 329 jointly known tasks; its 95% task-bootstrap interval is $[-2.43, 1.52]$ points

Support	Succ.	%	vs. bias	p	vs. prev.
GPT-5.5					
bias only	247 (1)	74.4	—	—	—
1/3 (106)	251	75.6	4/1	0.38	4/1
2/3 (212)	256	77.1	9/1	0.021	5/0
3/3 (318)	258	77.7	12/2	0.013	3/1
DeepSeek					
bias only	242 (3)	72.9	—	—	—
1/3 (106)	240 (3)	72.3	14/16	0.86	14/16
2/3 (212)	240 (3)	72.3	2/4	0.69	17/17
3/3 (318)	244 (3)	73.5	2/0	0.50	6/2

Table 38: Original executions of KeyOpt refitted on growing shares of the 318 support tasks (number of tasks in parentheses; Succ. gives unknowns in parentheses), evaluated on all 332 ToolQA tasks without the reranker at fixed residual scale 2. Paired columns compare each stage with bias only and with the previous stage.

(10,000 draws, seed 17). The shared-physical-unknown bound over all 332 tasks is $[-2, -1]$ net tasks; this is an identification bound, not a confidence interval. Here source-group IDs equal task IDs, so resampling is not template- or domain-clustered. These offset-transfer tests exclude both hypergraph recall and cited review.

Reranker ablation. Table 41 crosses the frozen reranker with the KeyOpt residual weight on the calibrated index and repeats the component ablation without the reranker; the reranker-free full system stays above every ToolQA baseline. Index rows hold the compact Top-10, support fits, documents and budgets fixed; only the scorer changes. Without the reranker, review adds 4 and 5 successes to the index (5/1 and 6/1 rescues/harms with GPT-5.5 and DeepSeek), and combining it with hypergraph recall adds 8 and 9 over recall alone (9/1, $p = 0.021$; 14/5). Against the reranker-free index, Full gives 8/1 rescues/harms with GPT-5.5 ($p = 0.04$) and 12/3 with DeepSeek ($p = 0.035$).

I.9 A Conditional Skill Boundary

We examine a narrower question than end-to-end Full improvement: can a verified execution issue reveal a condition under which the preferred skill changes, and can the existing 512-d space represent that condition? This retrospective diagnostic uses Coffee support tasks, not the 332 evaluation tasks; its teacher was designed after inspection.

Local intervention. Nineteen trajectories (14 GPT, five DeepSeek; 14 distinct tasks across models) use the single-day guide and encounter an empty result when filtering another date after an earlier date filter. Replaying the recorded Table-tool prefix reproduces each return exactly. Four fixed continuations give: original-state filtering, 19/19 errors; reloading the Coffee table before the same filter, 19/19 nonempty target results; retrying without reloading, 19/19 errors; and reloading before a synthetic absent-date query, 19/19 errors. These 76 local branches establish a table-state effect on the immediate return of the second filter.

Condition-dependent preference. We retain all 82 support tasks whose two executed candidates are exactly the single-day and range guides. The public-text teacher chooses the range guide when the question contains at least two distinct literal YYYY-MM-DD dates and the single-day guide otherwise. Table 42 reports both strata.

Compilation and limits. Three source-disjoint folds fit opposite corrections for the two keys using training questions and teacher labels, with a fixed ridge objective; task rewards do not enter this fit. Test-time inference uses only the existing query embedding and the corrected keys, without the date rule. The compiled choices match the teacher on all 82 held-out questions. Reusing each selected branch's observed outcome gives the counts in Table 42; relative to original local KeyOpt, GPT has seven rescues and four harms, and DeepSeek seven rescues, three harms and one unknown pair. The known-success increases come from the fewer-literal-date stratum.

The shuffled teacher uses the same objective, but its fitted key norms are larger (2.13–2.40 versus 0.574–0.613), preventing a clean effect-size-matched attribution. Both exceed the original fitting radii, and the rule and original KeyOpt use different supervision. The probe shows that this conditional boundary is expressible in the compressed space; extracting such boundaries automatically from model outputs is left to future work.

I.10 Review Decisions with Fixed Strengths

Figure 8 breaks down ToolQA outcomes by domain. Figure 14d groups outcomes by the number of events a reviewer cites per task.

Where events add to review. Events enter review in two ways: recall can add skills outside the semantic pool, and reading can reorder which eight candidates are reviewed. We split tasks by whether the eight reviewed candidates differ between review with and without events, and compare the two

A Indexing and calibration (transferred)

Configuration	DeepSeek Succ.	DeepSeek R/H	GPT-5.5 Succ.	GPT-5.5 R/H
No Skills	200		212	
BM25	226		237	
Dense, 4096-d	234		245	
(1) Compact, 512-d	232		244	
(1) + reliability bias	233		241	
+ KeyOpt (2)	240	7/0	242	1/0
Reranker + bias	239		246	
+ KeyOpt (2)	247	10/2	255	11/2
(1)+(2) + doc. review	256	16/0	264	22/0
+ execution history	257		265	

B Components (own feedback, GPT-5.5, reranker)

Configuration	Succ.	Hit@1	MRR	Cmp.	Worst	R/H
Index (1)+(2)	249	53.6	67.1	63.5	60.0	—
+ KeyOpt $\times 1$	253	56.0	68.5	65.2	61.4	4/0
+ KeyOpt $\times 2$	257	59.9	70.8	66.3	62.9	9/1
+ HG reading	251	53.9	—	64.6	60.0	2/0
+ read + recall	251	46.7	—	64.6	61.4	4/2
+ doc. review	262	79.5	—	66.3	67.1	16/3
CompSkill (all)	**264**	**79.8**	—	68.0	68.6	17/2

R/H: in A, KeyOpt against the preceding bias row and review against (1)+KeyOpt; in B, every row against the index. MRR@10; Cmp.: 178 complex queries; Worst: lowest domain. KeyOpt $\times 1$ and $\times 2$: scale fixed without the validation gate.

Table 39: ToolQA ablations ($n = 332$): **(A)** transferred Qwen3-8B feedback; **(B)** GPT-5.5's own support feedback, with the frozen reranker. Success counts and paired rescues/harms (R/H) use all 332 tasks.

Target executor	None	GPT source	DS source
GPT-5.5	241 (0)	249 (0)	252 (0)
DeepSeek	230 (2)	236 (2)	237 (3)

Table 40: Frozen calibration transfer: successes/332 (unknowns). Sources transfer reliability offsets; all rows use compact retrieval and reranking. Original execution records; with completed outcome accounting, DeepSeek reaches 232/238/240 (Figure 21, Table 28).

Scorer	Residual	GPT-5.5	DeepSeek
Calibrated index (compact Top-10, no hypergraph)			
with reranker r	bias only	249 (0)	240 (0)
	$\times 1$	253 (0)	238 (3)
	$\times 2$	257 (0)	239 (3)
without r	bias only	248 (0)	242 (3)
	$\times 1$	251 (0)	242 (3)
	$\times 2$	258 (0)	244 (3)
Components with r (default scorer)			
index	selected	249 (0)	240 (0)
+ hypergraph only	selected	251 (0)	247 (0)
+ review only	selected	262 (0)	252 (0)
full system	selected	264 (0)	**259 (0)**
Components without r (residual $\times 2$; robustness)			
index	$\times 2$	258 (0)	244 (3)
+ hypergraph only	$\times 2$	257 (0)	244 (3)
+ review only	$\times 2$	262 (0)	249 (3)
full system	$\times 2$	**265 (0)**	256 (0)

Table 41: Component and reranker ablations on ToolQA: successes/332 (remaining unknowns). Residual weights $\times 1/\times 2$ deploy the fitted residual at fixed weights.

Literal dates (n)	GPT Single	GPT Range	DeepSeek Single	DeepSeek Range
Fewer than two (41)	37	33	36^1	32^1
Two or more (41)	17	34	3^1	31
Selection on all 82 tasks				
	GPT		DeepSeek	
Best fixed guide	67		63^1	
Frozen reranker	63		60^1	
Original local KeyOpt	68		63^1	
Conditional teacher / key	71		67^1	
Shuffled-teacher key	58		50^2	

Table 42: Known successes in the completed two-branch support matrix; superscripts count unknown outcomes. Single/range denote the two Coffee guides. Outcomes come from completed support executions.

ToolQA, full system	GPT-5.5	DeepSeek
Reviews	332	332
Valid outputs	321	321
Fallback to proposal	11	11
Proposal kept	186	189
Proposal switched	146	143
Switched: rescued/harmed	15/2	19/7
Reviews citing events	190	265
Events cited per review	1.05	1.63
Input tokens per review	13.1K	11.5K
Output tokens per review	168	1644
Median seconds per review	10.7	5.0

Table 43: Cited review of the default CompSkill (frozen reranker) on all 332 ToolQA tasks. Outcomes of switched choices are paired against the proposal.

arms' outcomes (Table 44). Almost all of the event gain lies in tasks whose candidates change; when the labeled guide enters only with events, review rescues seven tasks and harms none.

Table 43 and Figure 14 characterize review. Both reviewers return a valid, quoted choice in more than 96% of tasks and switch about two in five proposals. GPT-5.5 keeps 186 proposals and switches

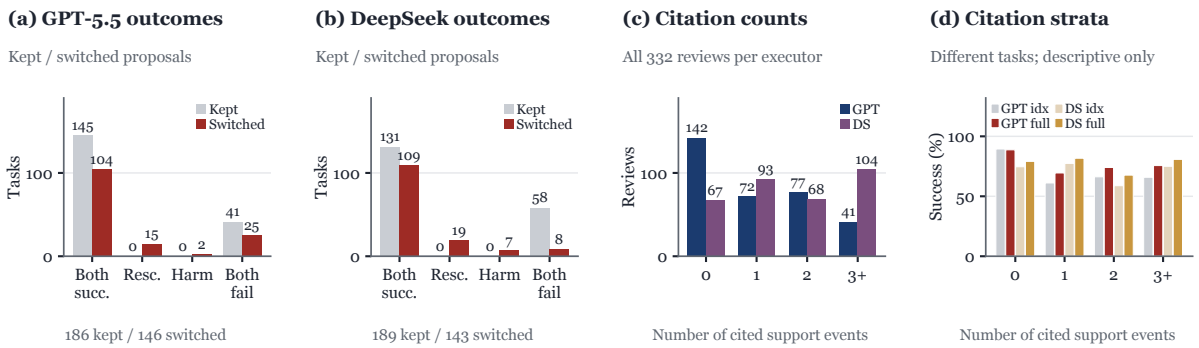


Figure 14: Review diagnostics on 332 ToolQA tasks: (a, b) outcomes against the hypergraph proposal, (c) citation counts and (d) success by citation count. Strata contain different tasks.

Tasks	GPT	DS	Pooled
Changed by events	5/2 (149)	7/1 (154)	12/3
Unchanged	0/1 (183)	2/1 (178)	2/2
Labeled guide added	3/0 (6)	4/0 (11)	7/0

Table 44: Rescues/harms of review with events over review without events, by whether events change the eight reviewed candidates (tasks in parentheses). Changed stratum: net $+9$ tasks, task-clustered bootstrap 95% CI $[1, 18]$ (10,000 draws resampling task IDs with both executors kept together, seed 17); per executor, exact McNemar $p = 0.45$ and 0.070, respectively.

146; switches rescue 15 tasks and harm two, versus 19/7 with DeepSeek. Kept proposals share the same execution outcome. DeepSeek cites events more often than GPT-5.5 (1.63 versus 1.05 per review) and spends more output tokens, while the input cost of about 12–13K tokens per review is dominated by the eight candidate documents. Earlier online variants are recorded in Appendix I.13. `artifact_manifest.json` links selections, reviewer exchanges, execution receipts, official scores and SHA-256 content hashes.

What the hypergraph exposes. Figure 15 counts the evidence that reaches review. Almost every task receives events: the reviewer sees three events for 297 tasks with GPT-5.5 and 330 with DeepSeek, so the cap of six is never binding, because one event per contrasting arm pair is kept. Recall changes the reviewed candidates for 114 tasks with GPT-5.5 and 147 with DeepSeek, adding up to four skills that compact keys rank outside their Top-10. These are the only routes through which experience can introduce a document the semantic index never proposed for that task.

How review rescues tasks. Table 45 lists rescues across domains and executors. Two patterns dominate. In Flights, review replaces the aggregation guide with the lookup guide for single-record questions, quoting the guide's own scope statement. In Yelp and Airbnb, it replaces a web skill or a guide from another domain with the database guide of the task's domain, citing events in which that guide succeeded on similar questions. In every rescue the quote locates the passage matching the question type, so one sentence verifies the decision.

I.11 Selection Changes Behind the Gains

The ToolQA library contains eight local database guides and 777 other documents. To distinguish recovery of a domain guide from choosing between its procedures, we classify all 332 tasks for each executor and four single-document references by their selected IDs. We use four categories: switching between the two guides of the task's database; entering that domain-guide set; leaving it; and all other or unchanged selections. Classification uses the public guide–domain mapping, not the task's outcome or the annotated best guide. All comparisons use the main table's outcome accounting, including remaining unknowns.

Against the calibrated index, all 17 GPT rescues occur within a database (one harm there, one elsewhere). DeepSeek has 15 rescues/four harms within a database, eight/one when entering its guide set, and one/zero when leaving it. Thus the gains are not confined to rejecting off-domain documents. Against document review without events, the within-domain changes are smaller: three/two for GPT and six/one for DeepSeek. Against Vector Top-1 they are 14/one and 17/one; GPT additionally has one unknown pair in that stratum. Each stratum groups supplied-document transitions of the Full-versus-index comparison, in which recall, reading and review change together; multi-

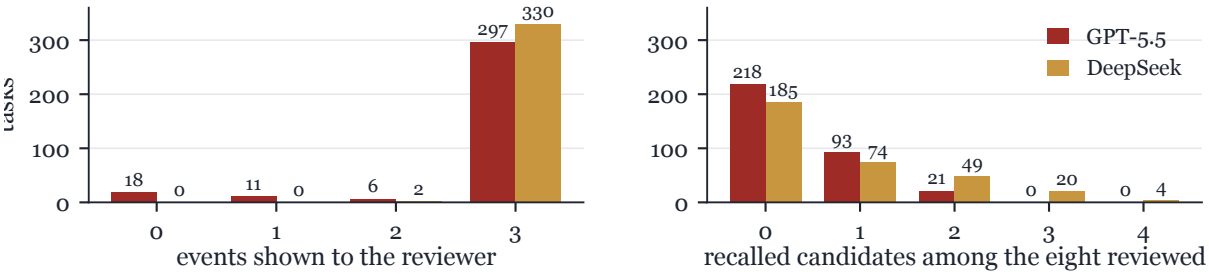


Figure 15: What the hypergraph exposes on 332 ToolQA tasks: the number of events shown to the reviewer, and the number of recalled skills from outside the semantic pool among the eight reviewed candidates.

Task	Exec.	Proposal → review choice	Cites	Quote from the chosen document
00342	GPT	Flights aggregation → Airbnb aggregation	3	`Proportions. Use FilterDB cumulatively`
00859	DS	Flights aggregation → Flights lookup	1	`For any question requiring a single flight record:`
00881	DS	Flights aggregation → Flights lookup	2	`'ActualElapsedTime' | Total gate-to-gate time in minutes`
01328	DS	web skill legalzoom → Yelp lookup	1	`Report coordinates in the order 'latitude, longitude' as ...`
01428	DS	web skill legalzoom → Yelp aggregation	0	`SELECT postal_code, COUNT(*) AS cnt FROM yelp_data WHERE ...`

Table 45: Rescues by review on ToolQA under the earlier fixed-strength calibration: the hypergraph proposal fails, review switches to another document and the executor succeeds. Quotes are verbatim and abridged.

Full relative to	Within domain	Into domain	Out of domain	Other / same	Total	Unknown
GPT-5.5						
Index + calibration	17/1	0/0	0/1	0/0	17/2	0
Review, no events	3/2	0/0	2/1	0/0	5/3	0
Vector Top-1	14/1	6/1	0/1	1/2	21/5	1
SkillRouter	16/0	14/2	0/1	0/1	30/4	0
DeepSeek						
Index + calibration	15/4	8/1	1/0	0/0	24/5	0
Review, no events	6/1	2/0	1/1	0/0	9/2	0
Vector Top-1	17/1	8/3	0/0	2/0	27/4	0
SkillRouter	17/0	20/6	0/0	3/1	40/7	0

Table 46: Rescues/harms of Full, grouped by the change in supplied document. Each row covers all 332 tasks, including unchanged outcomes; unknown: pairs with an unknown outcome.

document references are excluded because a bundle also changes the amount of context.

How the components work together. Compact keys establish which procedures are available for selection. KeyOpt uses matched outcomes to adjust preferences within that pool: the offset captures a skill's reliability across requests, while the residual can express preferences that vary with the query. Neither can promote a document absent from the pool. Events retain the conditions and alternatives behind the feedback, allowing recall to add such a document and review to assess its applicability. Table 5 illustrates the missing-candidate case; the Flights example below illustrates an available candidate whose scope better matches the request. The modules thus address different selection errors while keeping the supplied documents fixed.

I.12 Additional Case Studies

Review corrects calibration (ToolQA 00830, Flights)

Task. *What was the departure time of the AA2319 flight from MIA to LAS on 2022-06-05?*

Index and hypergraph. Both propose the multi-record aggregation guide for US flight data; GPT-5.5 and DeepSeek both fail the task with it.

Events read. Three same-type support tasks (departure times of DL891 and DL5172, and a CRS-versus-actual delay), in each of which the single-record lookup guide succeeded and a web flight skill failed.

Review. Selects the single-record lookup guide, quoting `"For any question requiring a single flight record:"` and citing two of the events.

Outcome. CompSkill succeeds with both executors, as do Vector Top-1, SkillRouter and review without events, which all supply the same guide as CompSkill.

Calibration favored the aggregation guide because it wins on the many multi-record flight questions among the support tasks. The query of this task is a single-record lookup, and the cited events show the lookup guide succeeding on exactly this question type. Review thus repairs a global preference with local, attributable evidence, the division of labor between KeyOpt and the hypergraph that Section 4.7 measures in aggregate.

Recall finds a DeepSeek-only guide (ToolQA 00988)

Task. *How many flights from ITO to LAX host by United Air Lines Inc.?*

Index. Supplies a web flight-search skill; DeepSeek fails, as it does with every baseline.

Recall. Reading a support event in which the multi-record aggregation guide answered a route count (*How many flights from ATW to ORD host by Air Wisconsin Airlines Corp?*) brings this guide into the candidate pool and then to the top.

Review. Keeps it, quoting `"When counting how many distinct flight services operate a given route under a given airline, count unique values of 'Flight_Number_Marketing_Airline', not total rows."` and citing that event.

Outcome. DeepSeek succeeds; review without events chooses the lookup guide and fails. The same guide fails with GPT-5.5, so its value is executor-specific.

This case isolates recall: the guide enters only through an event recorded with DeepSeek, and it helps DeepSeek but not GPT-5.5. Utility is executor-specific, which is why events are read only by the executor that produced them and why CompSkill keeps per-executor calibration.

I.13 Earlier Configurations

Table 47 and Figure 16 collect earlier configurations with their original task counts, executors and protocols. They are separate records from the frozen configurations in Table 25.

J Additional Diagnostics for RQ2–RQ7

These analyses reuse completed predictions, selections and outcomes without new model calls.

RQ2: Scope of the component controls. Figure 17 expands Table 28. Removing learned key construction means using the same encoder's first 512 coordinates with normalization, rather than replacing the encoder. The calibration comparison freezes compact retrieval and the reranker, then adds offsets learned from the target executor's own support executions. The residual is analyzed in Section 4.6. The final comparison removes the combined event-reading, candidate-recall and document-review interface from Full.

RQ3: Recovering the last missing skill. For query q with annotated relevant set G_q, define $r_q = \max_{s \in G_q} \mathrm{rank}_q(s)$. Completeness@10 is exactly $r_q \leq 10$. We place r_q in one of four bands: 1–3, 4–10, 11–15 and > 15; skills absent from the saved Top-100 are included in the last band. Each cell of Figure 18 counts queries with the corresponding native and compact ranks, so improvements and regressions are both visible. Sorting each saved prediction reproduces its previously scored Recall@10 value exactly.

Across all 4,392 queries, seeds 17/29/43 newly complete 492/500/467 sets and lose 123/104/136, for net gains of 369/396/331. On the 2,249 multi-skill queries, the corresponding gains/losses are 350/98, 361/82 and 315/105: net 252/279/210. Thus the aggregate completeness improvement is present in every seed and includes retrieval of the last required document. The unlearned truncation and projection results remain in Tables 14 and 15.

RQ4: From score edits to executed choices. Figure 19 keeps all 3,320 candidate scores from the fixed-scale GPT diagnostic. Of these, 708 change; 39 tasks change their first choice and 293 do not. The 39 changed choices yield nine rescues, one harm and 29 unchanged outcomes. The residual thus acts selectively: most edits leave the first choice intact, and changed choices rescue nine times as often as they harm. It is evaluated at scale 2 with the original candidate set and reranker held fixed; the default uses validation-selected calibration. Appendix I.8 retains all scales, executor differences and the original paired test. Appendix I.9 describes the cross-fitted conditional-choice analysis and its teacher assumptions.

RQ5: What changes when experience is used? Figure 20 decomposes every calibrated-index–Full comparison using the supplied document ID and public database mapping. GPT keeps the document on 195 tasks and switches on 137; DeepSeek keeps it on 200 and switches on 132. Same-document strata keep their outcomes by construction.

GPT's 69 within-database switches contain 17 rescues, one harm and 51 unchanged outcomes. DeepSeek's 57 contain 15 rescues, four harms and

Study	Executor	n	Results
AppWorld Challenge, online KeyOpt	DeepSeek / GPT-5.5	90	offline 74 / 81; online 69 / 79 (2 unknown)
ALFWorld, earlier version, online	DeepSeek	134	offline 121; online 119
ToolQA, SRA-specific compressor	DeepSeek	332	No Skills 202; compact 257; +bias 260; +KeyOpt 259; +doc. 266 (bias) / 267 (KeyOpt); +hist. 269
ToolQA, SRA-specific compressor	Qwen3-8B	332	No Skills 147; BM25 166; dense 4096-d 190; compact 219; +bias 224; +KeyOpt 223; +doc. 227 / 225; +hist. 224
ToolQA, full-library SkillRouter	DeepSeek	332	230 (241 when reranking the shared Top-10)
ToolQA, fixed-scale KeyOpt residual	GPT-5.5	332	bias 249; residual $\times 1$: 253 (4/0); $\times 2$: 257 (9/1)
ALFWorld, dynamic-state protocol	DeepSeek	134	No Skills 76; compact 96; +bias 100; +doc. 115
SkillsBench	DeepSeek	60	GoSkills 25; dense 26; compact+bias 25; +KeyOpt 25; +group evidence 25

Table 47: Additional studies (successes / n). "Doc.": document review; "hist.": history review.

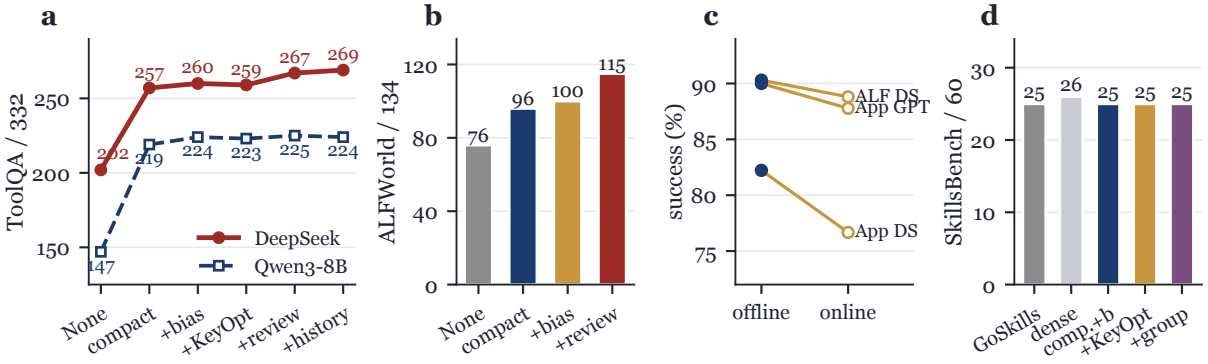
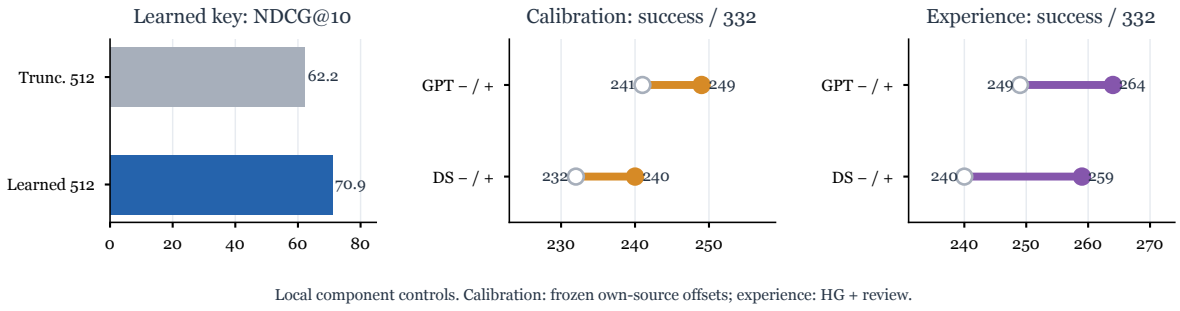


Figure 16: Earlier configurations from Table 47. (a) ToolQA ladders with an SRA-specific compressor. (b) ALFWorld with a dynamic-state protocol. (c) Offline versus online KeyOpt. (d) SkillsBench with DeepSeek.



Local component controls. Calibration: frozen own-source offsets; experience: HG + review.

Figure 17: Three local component controls. Open/filled points denote removal/enabling. SkillRet shows NDCG@10; ToolQA shows successes/332. DeepSeek calibration gives 232/240 without/with offsets.

38 unchanged outcomes. Of its remaining nine rescues, eight switch into a task's database guide and one switches away from such a guide; database membership alone is not a utility label. All 332 tasks per executor, including unchanged outcomes and harms, remain in the flow. The full reference-by-reference decomposition appears in Table 46.

The cached 2×2 comparison separates event use from document review: GPT reaches 249 without either, 251 with events, 262 with review and 264 with both; DeepSeek reaches 240, 247, 252 and 259. Event use denotes both reading and candidate recall. The four cells measure the contribution of each interface conditional on the other.

RQ6: Two distinct transfer tests. The left panel of Figure 21 transfers the frozen compressor to different libraries; it reports mean Recall@10 over the original three seeds. SRA has 665 queries, including the 200-query ToolQA subset; these are overlapping samples, not two independent benchmarks. The AppWorld API analysis has 57 cached requests and is separate from the 90-task execution

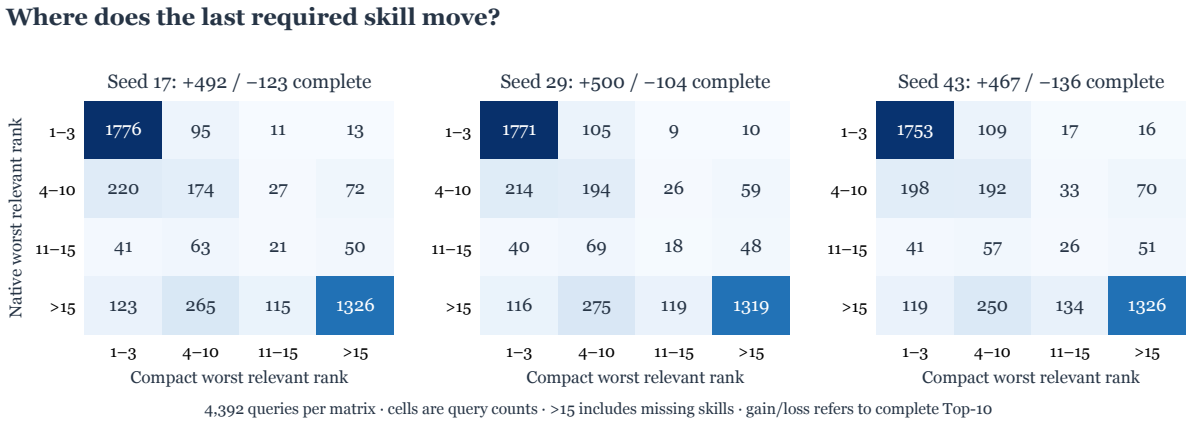


Figure 18: Movement of the worst-ranked required skill for every SkillRet TEST query and each compact seed: rows are native ranks, columns compact ranks, and panel titles count Top-10 sets that gain or lose completeness.

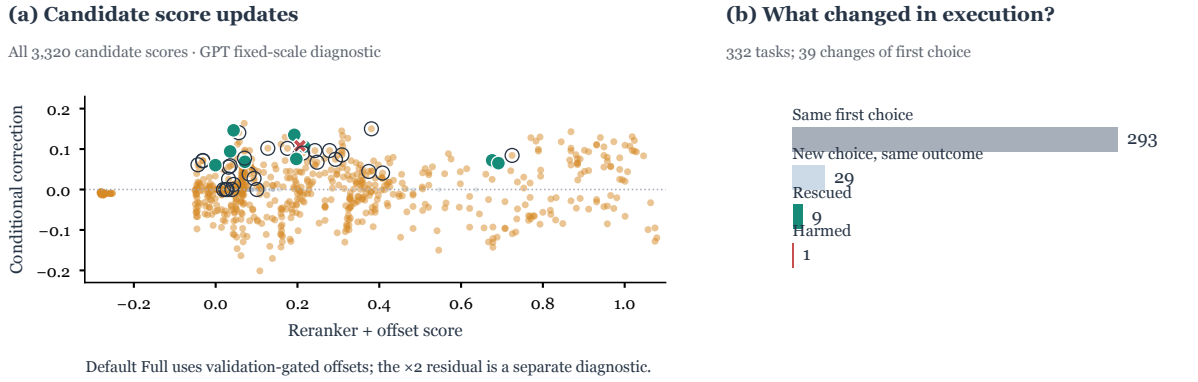


Figure 19: KeyOpt edits and their executed consequences. Gray points are unmodified scores and amber points are modified scores. Green circles mark rescued first choices, the red cross marks the harmed choice, and open circles mark changed choices with unchanged outcomes. The right panel retains all 332 tasks.

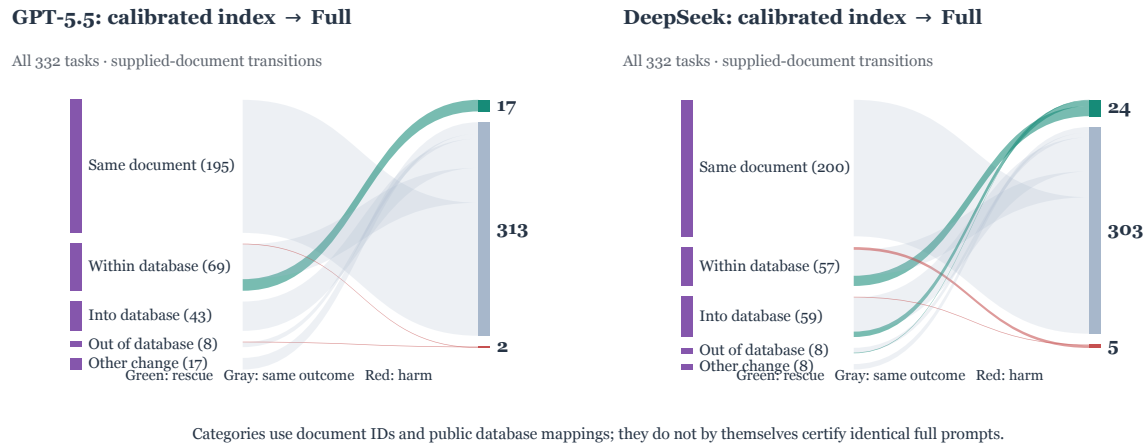


Figure 20: All-task document transitions from calibrated index to Full. Ribbons are proportional to task counts; green denotes a rescue, red a harm, and gray an unchanged outcome. Categories describe the relation between supplied guides and the task's database, without attributing the change to an individual component.

evaluation. Native/compact values are 71.85/74.41, 43.00/60.00 and 36.27/38.05. Recall@1 and all seeds remain in Appendix F.3; the table reports both first-choice and candidate-coverage metrics.

The right panel transfers frozen offsets across executors with no target-executor refitting. Against no calibration, DeepSeek-source offsets add 11 known successes for GPT; GPT-source offsets add

six for DeepSeek. Own-source offsets add eight successes for each of the two executors.

RQ7: Cost of the index interface. Figure 22 uses the saved 1,000-key microbenchmark, with 20 warm-up and 200 timed cached queries on two CPU threads. It includes a dot product and stable Top-10 sorting, excluding encoding, reranking, review and execution. Native median/P90 latency is 0.21060/0.24480 ms; compact latency is 0.06014/0.07723 ms. This gives a 3.50-fold median reduction in lookup time for this workload.

Float32 base keys occupy 16.384 versus 2.048 MB. Allowing a full 512-d residual and one offset per skill gives $1{,}000(512 + 512 + 1)4 = 4{,}100{,}000$ bytes, about one quarter of the native base-key array. This is a parameter count, not measured peak RAM; event storage and model weights are excluded. The reviewer adds one call: the cached ToolQA reference uses about 13.1K/11.5K input tokens per GPT/DeepSeek review and median 10.7/5.0 s (Table 43). The claim is a cheaper retrieval index with an explicit review cost, not that Full is cheaper than a selector without review.

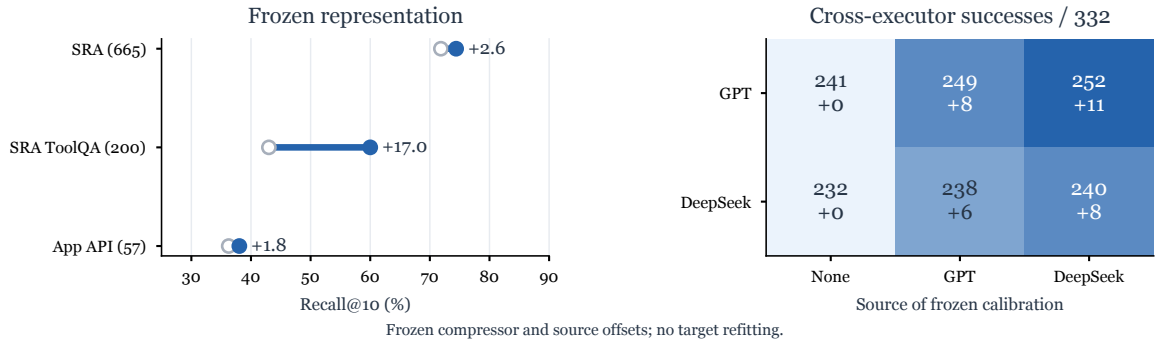


Figure 21: Frozen transfer across libraries and executors. Open/filled points denote native/compact representations. The matrix shows successes/332 and changes from no calibration, using the completed outcome accounting. The two transfer tests evaluate different parameters: the compressor and the calibration offsets.

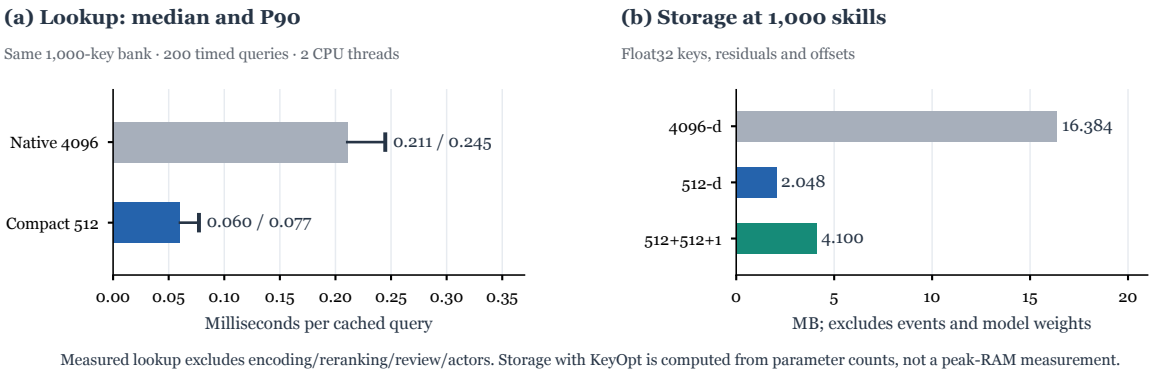


Figure 22: Measured cached lookup and analytical storage. Black ticks mark P90, not uncertainty. The last storage bar adds one residual vector and offset per skill and excludes events, model weights and allocator overhead.